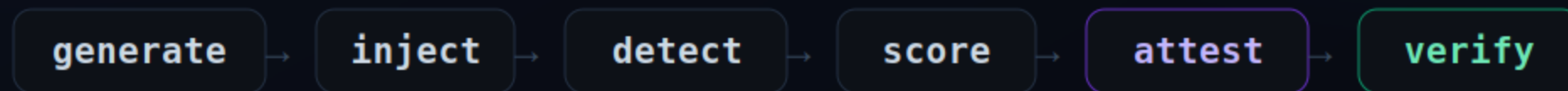


ProvingGround

Proving market-surveillance models work — by manufacturing the labelled reality they're missing.



● THE PROBLEM

Detection is easy to claim.
It's impossible to **prove**.

Regulated firms must show their surveillance models catch abuse.

- Real trading data **isn't labelled** — nobody tags which trades were spoofing.
- So there's **no ground truth** to measure a detector against.
- The hard problem isn't detection — it's **validation**. That's the gap.

● THE IDEA

Manufacture the labelled reality.

Generate realistic order books, inject the spoofing ourselves, and record exactly which events are manipulation. Now detection is measurable.

KNOWN BY CONSTRUCTION

Ground truth, for free

Every scenario is labelled clean / manipulated, with the implicated spoof event IDs.

TAMPER-EVIDENT

Provable, not asserted

Each result is written to Walrus and re-verifiable by recomputing its hash.

HEAD-TO-HEAD

Two detectors, scored

Heuristic vs finance-tuned LLM — catch rate, false positives, and the exact misses.

SPEC-DRIVEN

Built with Codeplain

The whole core is generated from one spec — plus its conformance tests.

● HOW IT WORKS

One deterministic loop, end to end.

- **generate / inject** — clean stochastic order flow, then a planted spoof
- (large order far from mid, cancelled before fill) with labelled ground truth.
- **detect / score** — both detectors; catch rate, FPR, precision, and a
- drill-down into every miss (order book + why it was manipulation).
- **attest / verify** — write the run to Walrus; verify by recomputing the
- hash. Tamper a byte and verification fails.
- **calibrated to a real market** — the live DeepBook SUI/USDC book seeds realism.

```
python proving_ground.py benchmark --seed 7 --count 10
```

The same loop satisfies every track.

CODEPLAIN

Spec -> code -> tests

12 specs, 68 conformance tests,
19/100 credits. We generate tests for
models; they generate tests for code.

SUI — WALRUS + DEEPBOOK

Verifiable + real

Tamper-evident record on Walrus;
live DeepBook SUI/USDC as the
market reference.

FLOCK — SOVEREIGN AI

Finance-native LLM

qwen3-235b-a22b-thinking-qwfin,
nano-USD cost. Surveillance is exactly
what you can't outsource.

BGA · SOLVIMON

Fair + fundable

Benchmark integrity is checkable;
metered per eval-run, with a buyer
obligated to validate.

● STATUS & WHAT'S NEXT

Demo-ready today.

68

conformance tests passing

39

unit tests passing

19

/ 100 Codeplain credits

- **Proven live:** Walrus write -> verify -> tamper-detection; DeepBook reads; the loop.
- **Next:** FLock account top-up for the live LLM head-to-head (code is mock-proven);
- Solvimon key to flip metering live; Vercel deploy (core as a service).
- **Vision:** validation infrastructure for the firms obligated to prove their models.